Quiz Questions

Q1: Operating Systems (1)
What is an operating system?
A. interface between the hardware and application programs
B. collection of programs that manages hardware resources
C. system service provider to the application programs
D. All the options
Correct Answer: D. All the options

Q2: Operating Systems (10)
Which one of the following is not a real time operating system?
A. RTLinux
B. Palm OS
C. QNX
D. VxWorks
Correct Answer: B. Palm OS

Q3: Operating Systems (100)
In the non blocking send _____
A. the sending process keeps sending until the message is received
B. the sending process sends the message and resumes operation
C. the sending process keeps sending until it receives a message
D. none
Correct Answer: B. the sending process sends the message and resumes operation

Q4: Operating Systems (101)
In the Zero capacity queue _____
A. the queue can store at least one message
B. the sender blocks until the receiver receives the message
C. the sender keeps sending and the messages don’t wait in the queue
D. none
Correct Answer: B. the sender blocks until the receiver receives the message

Q5: Operating Systems (102)
The Zero Capacity queue _____
A. is referred to as a message system with buffering
B. is referred to as a message system with no buffering
C. is referred to as a link
D. none
Correct Answer: B. is referred to as a message system with no buffering

Q6: Operating Systems (103)
Bounded capacity and Unbounded capacity queues are referred to as _____
A. Programmed buffering
B. Automatic buffering
C. User defined buffering
D. No buffering
Correct Answer: B. Automatic buffering

Q7: Operating Systems (104)
A process that is based on IPC mechanism which executes on different systems and can communicate with other processes using message based communication, is called _____
A. Local Procedure Call
B. Inter Process Communication
C. Remote Procedure Call
D. Remote Machine Invocation
Correct Answer: C. Remote Procedure Call

Q8: Operating Systems (105)
The initial program that is run when the computer is powered up is called _____
A. boot program
B. bootloader
C. initializer
D. bootstrap program
Correct Answer: D. bootstrap program

Q9: Operating Systems (106)
What is a trap/exception?
A. hardware generated interrupt caused by an error
B. software generated interrupt caused by an error
C. user generated interrupt caused by an error
D. none
Correct Answer: B. software generated interrupt caused by an error

Q10: Operating Systems (107)
In a memory mapped input/output _____
A. the CPU uses polling to watch the control bit constantly, looping to see if a device is ready
B. the CPU writes one data byte to the data register and sets a bit in control register to show that a byte is available

C. the CPU receives an interrupt when the device is ready for the next byte
D. the CPU runs a user written code and does accordingly
Correct Answer: B. the CPU writes one data byte to the data register and sets a bit in control register to show that a byte is available

Q11: Operating Systems (108)
Which module gives control of the CPU to the process selected by the short-term scheduler?
A. dispatcher
B. interrupt
C. scheduler
D. none
Correct Answer: A. dispatcher

Q12: Operating Systems (109)
The processes that are residing in main memory and are ready and waiting to execute are kept on a list called _____
A. job queue
B. ready queue
C. execution queue
D. process queue
Correct Answer: B. ready queue

Q13: Operating Systems (11)
What does OS X has?
A. monolithic kernel with modules
B. microkernel
C. monolithic kernel
D. hybrid kernel
Correct Answer: D. hybrid kernel

Q14: Operating Systems (110)
The interval from the time of submission of a process to the time of completion is termed as _____
A. waiting time
B. turnaround time
C. response time
D. throughput
Correct Answer: B. turnaround time

Q15: Operating Systems (111)

Which scheduling algorithm allocates the CPU first to the process that requests the CPU first?
A. first-come, first-served scheduling
B. shortest job scheduling
C. priority scheduling
D. none
Correct Answer: A. first-come, first-served scheduling

Q16: Operating Systems (112)
In priority scheduling algorithm _____
A. CPU is allocated to the process with highest priority
B. CPU is allocated to the process with lowest priority
C. Equal priority processes can not be scheduled
D. None
Correct Answer: A. CPU is allocated to the process with highest priority

Q17: Operating Systems (113)
In priority scheduling algorithm, when a process arrives at the ready queue, its priority is compared with the priority of _____
A. all process
B. currently running process
C. parent process
D. init process
Correct Answer: B. currently running process

Q18: Operating Systems (114)
Which algorithm is defined in Time quantum?
A. shortest job scheduling algorithm
B. round robin scheduling algorithm
C. priority scheduling algorithm
D. multilevel queue scheduling algorithm
Correct Answer: B. round robin scheduling algorithm

Q19: Operating Systems (115)
Process are classified into different groups in _____
A. shortest job scheduling algorithm
B. round robin scheduling algorithm
C. priority scheduling algorithm
D. multilevel queue scheduling algorithm
Correct Answer: D. multilevel queue scheduling algorithm

Q20: Operating Systems (116)
In multilevel feedback scheduling algorithm _____
A. a process can move to a different classified ready queue
B. classification of ready queue is permanent
C. processes are not classified into groups
D. none
Correct Answer: A. a process can move to a different classified ready queue

Q21: Operating Systems (117)
Which one of the following can not be scheduled by the kernel?
A. kernel level thread
B. user level thread
C. process
D. none
Correct Answer: B. user level thread

Q22: Operating Systems (118)
CPU scheduling is the basis of _____
A. multiprocessor systems
B. multiprogramming operating systems
C. larger memory sized systems
D. none
Correct Answer: B. multiprogramming operating systems

Q23: Operating Systems (119)
With multiprogramming _____ is used productively.
A. time
B. space
C. money
D. all
Correct Answer: A. time

Q24: Operating Systems (12)
In operating system, each process has its own _____
A. open files
B. pending alarms, signals, and signal handlers
C. address space and global variables
D. all
Correct Answer: D. all

Q25: Operating Systems (120)

What are the two steps of a process execution?
A. I/O & OS Burst
B. CPU & I/O Burst
C. Memory & I/O Burst
D. OS & Memory Burst
Correct Answer: B. CPU & I/O Burst

Q26: Operating Systems (121)
An I/O bound program will typically have _____
A. a few very short CPU bursts
B. many very short I/O bursts
C. many very short CPU bursts
D. a few very short I/O bursts
Correct Answer: C. many very short CPU bursts

Q27: Operating Systems (122)
A process is selected from the _____ queue by the _____ scheduler, to be executed.
A. blocked, short term
B. wait, long term
C. ready, short term
D. a few very short I/O bursts
Correct Answer: C. ready, short term

Q28: Operating Systems (123)
In the following cases non – preemptive scheduling occurs?
A. When a process switches from the running state to the ready state
B. When a process goes from the running state to the waiting state
C. When a process switches from the waiting state to the ready state
D. All
Correct Answer: B. When a process goes from the running state to the waiting state

Q29: Operating Systems (124)
The switching of the CPU from one process or thread to another is called _____
A. process switch
B. task switch
C. context switch
D. all
Correct Answer: D. all

Q30: Operating Systems (125)
What is Dispatch latency?

A. the speed of dispatching a process from running to the ready state
B. the time of dispatching a process from running to ready state and keeping the CPU idle
C. the time to stop one process and start running another one
D. none
Correct Answer: C. the time to stop one process and start running another one

Q31: Operating Systems (126)
Scheduling is done so as to _____
A. increase CPU utilization
B. decrease CPU utilization
C. keep the CPU more idle
D. none
Correct Answer: A. increase CPU utilization

Q32: Operating Systems (127)
What is Turnaround time?
A. the total waiting time for a process to finish execution
B. the total time spent in the ready queue
C. the total time spent in the running queue
D. the total time from the completion till the submission of a process
Correct Answer: D. the total time from the completion till the submission of a process

Q33: Operating Systems (128)
Scheduling is done so as to _____
A. increase the turnaround time
B. decrease the turnaround time
C. keep the turnaround time same
D. there is no relation between scheduling and turnaround time
Correct Answer: B. decrease the turnaround time

Q34: Operating Systems (129)
What is Waiting time?
A. the total time in the blocked and waiting queues
B. the total time spent in the ready queue
C. the total time spent in the running queue
D. the total time from the completion till the submission of a process
Correct Answer: B. the total time spent in the ready queue

Q35: Operating Systems (13)
In a timeshare operating system, when the time slot assigned to a process is completed, the process switches from the current state to?

A. Suspended state
B. Terminated state
C. Ready state
D. Blocked state
Correct Answer: C. Ready state

Q36: Operating Systems (130)
Scheduling is done so as to _____
A. increase the waiting time
B. keep the waiting time the same
C. decrease the waiting time
D. none
Correct Answer: C. decrease the waiting time

Q37: Operating Systems (131)
What is Response time?
A. the total time taken from the submission time till the completion time
B. the total time taken from the submission time till the first response is produced
C. the total time taken from submission time till the response is output
D. none
Correct Answer: B. the total time taken from the submission time till the first response is produced

Q38: Operating Systems (132)
Round robin scheduling falls under the category of _____
A. Non-preemptive scheduling
B. Preemptive scheduling
C. All
D. None
Correct Answer: B. Preemptive scheduling

Q39: Operating Systems (133)
With round robin scheduling algorithm in a time shared system _____
A. using very large time slices converts it into First come First served scheduling algorithm
B. using very small time slices converts it into First come First served scheduling algorithm
C. using extremely small time slices increases performance
D. using very small time slices converts it into Shortest Job First algorithm
Correct Answer: A. using very large time slices converts it into First come First served scheduling algorithm

Q40: Operating Systems (134)
The portion of the process scheduler in an operating system that dispatches processes is concerned with _____
A. assigning ready processes to CPU
B. assigning ready processes to waiting queue
C. assigning running processes to blocked queue
D. all
Correct Answer: A. assigning ready processes to CPU

Q41: Operating Systems (135)
What is FIFO algorithm?
A. first executes the job that came in last in the queue
B. first executes the job that came in first in the queue
C. first executes the job that needs minimal processor
D. first executes the job that has maximum processor needs
Correct Answer: B. first executes the job that came in first in the queue

Q42: Operating Systems (136)
The strategy of making processes that are logically runnable to be temporarily suspended is called _____
A. Non preemptive scheduling
B. Preemptive scheduling
C. Shortest job first
D. First come First served
Correct Answer: B. Preemptive scheduling

Q43: Operating Systems (137)
Complex scheduling algorithms _____
A. are very appropriate for very large computers
B. use minimal resources
C. use many resources
D. all
Correct Answer: A. are very appropriate for very large computers

Q44: Operating Systems (138)
What is FIFO algorithm?
A. first executes the job that came in last in the queue
B. first executes the job that came in first in the queue
C. first executes the job that needs minimal processor
D. first executes the job that has maximum processor needs

Correct Answer: B. first executes the job that came in first in the queue

Q45: Operating Systems (139)
The strategy of making processes that are logically runnable to be temporarily suspended is called _____
A. Non preemptive scheduling
B. Preemptive scheduling
C. Shortest job first
D. First come First served
Correct Answer: B. Preemptive scheduling

Q46: Operating Systems (14)
Cascading termination refers to the termination of all child processes if the parent process terminates _____
A. Normally or abnormally
B. Abnormally
C. Normally
D. None
Correct Answer: A. Normally or abnormally

Q47: Operating Systems (140)
There are 10 different processes running on a workstation. Idle processes are waiting for an input event in the input queue. Busy processes are scheduled with the Round-Robin time sharing method. Which out of the following quantum times is the best value for small response times, if the processes have a short runtime, e.g. less than 10ms?
A. tQ = 15ms
B. tQ = 40ms
C. tQ = 45ms
D. tQ = 50ms
Correct Answer: A. tQ = 15ms

Q48: Operating Systems (141)
Orders are processed in the sequence they arrive if _____ rule sequences the jobs.
A. earliest due date
B. slack time remaining
C. first come, first served
D. critical ratio
Correct Answer: C. first come, first served

Q49: Operating Systems (142)

Under multiprogramming, turnaround time for short jobs is usually _____ and that for long jobs is slightly _____
A. Lengthened; Shortened
B. Shortened; Lengthened
C. Shortened; Shortened
D. Shortened; Unchanged
Correct Answer: B. Shortened; Lengthened

Q50: Operating Systems (143)
Which of the following statements are true?
I. Shortest remaining time first scheduling may cause starvation
II. Preemptive scheduling may cause starvation
III. Round robin is better than FCFS in terms of response time
A. I Only
B. I and III only
C. II and III only
D. All
Correct Answer: D. All

Q51: Operating Systems (144)
The real difficulty with SJF in short term scheduling is _____
A. it is too good an algorithm
B. knowing the length of the next CPU request
C. it is too complex to understand
D. none
Correct Answer: B. knowing the length of the next CPU request

Q52: Operating Systems (145)
The FCFS algorithm is particularly troublesome for _____
A. time sharing systems
B. multiprogramming systems
C. multiprocessor systems
D. operating systems
Correct Answer: B. multiprogramming systems

Q53: Operating Systems (146)
Consider the following set of processes, the length of the CPU burst time given in milliseconds.

| Process | Burst time |
|---|---|
| P1 | 6 |
| P2 | 8 |
| P3 | 7 |

P4 3

Assuming the above process being scheduled with the SJF scheduling algorithm.

A. The waiting time for process P1 is 3ms
B. The waiting time for process P1 is 0ms
C. The waiting time for process P1 is 16ms
D. The waiting time for process P1 is 9ms
Correct Answer: A. The waiting time for process P1 is 3ms

Q54: Operating Systems (147)
An SJF algorithm is simply a priority algorithm where the priority is _____
A. the predicted next CPU burst
B. the inverse of the predicted next CPU burst
C. the current CPU burst
D. anything the user wants
Correct Answer: A. the predicted next CPU burst

Q55: Operating Systems (148)
A solution to the problem of indefinite blockage of low – priority processes is _____
A. Starvation
B. Wait queue
C. Ready queue
D. Aging
Correct Answer: D. Aging

Q56: Operating Systems (149)
Which of the following statements are true?
i) Shortest remaining time first scheduling may cause starvation
ii) Preemptive scheduling may cause starvation
iii) Round robin is better than FCFS in terms of response time
A. I only
B. I and III Only
C. II and III only
D. I, II, and III
Correct Answer: D. I, II, and III

Q57: Operating Systems (15)
When a process is in a "Blocked" state waiting for some I/O service. When the service is completed, it goes to the _____
A. Terminated state
B. Suspended state
C. Running state

D. Ready state
Correct Answer: D. Ready state

Q58: Operating Systems (150)
Mutual exclusion implies that _____
A. if a process is executing in its critical section, then no other process must be executing in their critical sections
B. if a process is executing in its critical section, then other processes must be executing in their critical sections
C. if a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution
D. none
Correct Answer: A. if a process is executing in its critical section, then no other process must be executing in their critical sections

Q59: Operating Systems (151)
Bounded waiting implies that there exists a bound on the number of times a process is allowed to enter its critical section _____
A. after a process has made a request to enter its critical section and before the request is granted
B. when another process is in its critical section
C. before a process has made a request to enter its critical section
D. none
Correct Answer: A. after a process has made a request to enter its critical section and before the request is granted

Q60: Operating Systems (152)
A minimum of _____ variable(s) is/are required to be shared between processes to solve the critical section problem.
A. 1
B. 2
C. 3
D. 4
Correct Answer: B. 2

Q61: Operating Systems (153)
The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as S0 = 1, S1 = 0, S2 = 0.

```
 Process P0
 while(true)
 {
    wait(S0);
```

```
    print '0';
    release(S1);
    release(S2);
}
Process P1
wait(S1);
release(S0);
Process P2
wait(S2);
release(S0);
```

How many times will P0 print ‘0’?

A. At least twice
B. Exactly twice
C. Exactly thrice
D. Exactly once

Correct Answer: A. At least twice

Q62: Operating Systems (154)

Each process Pi, i = 0,1,2,3,……,9 is coded as follows.

```
repeat
 P(mutex)
 {Critical Section}
 V(mutex)
 forever
```

The code for P10 is identical except that it uses V(mutex) instead of P(mutex). What is the largest number of processes that can be inside the critical section at any moment (the mutex being initialized to 1)?

A. 1
B. 2
C. 3
D. None

Correct Answer: C. 3

Q63: Operating Systems (155)

Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes.

```
Process P1 :
while(true)
{
w1 = true;
while(w2 == true);
Critical section
w1 = false;
}
```

Remainder Section
Process P2 :
while(true)
{
w2 = true;
while(w1 == true);
Critical section
w2 = false;
}
Remainder Section
Here, w1 and w2 have shared variables, which are initialized to false. Which one of the following statements is TRUE about the above construct?
A. It does not ensure mutual exclusion
B. It does not ensure bounded waiting
C. It requires that processes enter the critical section in strict alternation
D. It does not prevent deadlocks but ensures mutual exclusion
Correct Answer: D. It does not prevent deadlocks but ensures mutual exclusion

Q64: Operating Systems (156)
Semaphore is a/an _____ to solve the critical section problem.
A. after a particular process
B. periodically
C. atomically
D. none
Correct Answer: C. atomically

Q65: Operating Systems (157)
What are the two atomic operations permissible on semaphores?
A. wait and signal
B. stop and wait
C. hold and wait
D. none
Correct Answer: A. wait and signal

Q66: Operating Systems (158)
The following pair of processes share a common variable X.
Process A
int Y;
A1: Y = X*2;
A2: X = Y;
Process B
int Z;
B1: Z = X+1;

B2: X = Z;
X is set to 5 before either process begins execution. As usual, statements within a process are executed sequentially, but statements in process A may execute in any order with respect to statements in process B.
How many different values of X are possible after both processes finish executing?
A. 2
B. 3
C. 4
D. 8
Correct Answer: C. 4

Q67: Operating Systems (159)
The program follows to use a shared binary semaphore T.
Process A
int Y;
A1: Y = X*2;
A2: X = Y;
signal(T);
Process B
int Z;
B1: wait(T);
B2: Z = X+1;
X = Z;
T is set to 0 before either process begins execution and, as before, X is set to 5.
Now, how many different values of X are possible after both processes finish executing?
A. 1
B. 2
C. 3
D. 4
Correct Answer: A. 1

Q68: Operating Systems (16)
Transient operating system code is a code that _____
A. stays in the memory always
B. never enters the memory space
C. comes and goes as needed
D. is not easily accessible
Correct Answer: C. comes and goes as needed

Q69: Operating Systems (160)
All processes share a semaphore variable mutex, initialized to 1. Each process must execute wait(mutex) before entering the critical section and signal(mutex) afterward.
Suppose a process executes in the following manner.

signal(mutex);
.....
critical section
.....
wait(mutex);
In this situation :
A. a deadlock will occur
B. processes will starve to enter critical section
C. several processes maybe executing in their critical section
D. all
Correct Answer: C. several processes maybe executing in their critical section

Q70: Operating Systems (161)
All processes share a semaphore variable mutex, initialized to 1. Each process must execute wait(mutex) before entering the critical section and signal(mutex) afterward.
Suppose a process executes in the following manner.
wait(mutex);
.....
critical section
.....
wait(mutex);
A. a deadlock will occur
B. processes will starve to enter critical section
C. several processes maybe executing in their critical section
D. all
Correct Answer: A. a deadlock will occur

Q71: Operating Systems (162)
Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned.
Method used by P1 :
while(S1==S2);
Critical section
S1 = S2;
Method used by P2 :
while(S1!=S2);
Critical section
S2 = not(S1);
Which of the following statements describes properties achieved?
A. Mutual exclusion but not progress
B. Progress but not mutual exclusion
C. Neither mutual exclusion nor progress
D. Both mutual exclusion and progress

Correct Answer: D. Both mutual exclusion and progress

Q72: Operating Systems (163)
What is a reusable resource?
A. that can be used by one process at a time and is not depleted by that use
B. that can be used by more than one process at a time
C. that can be shared between various threads
D. none
Correct Answer: A. that can be used by one process at a time and is not depleted by that use

Q73: Operating Systems (164)
A system is in the safe state if _____
A. the system can allocate resources to each process in some order and still avoid a deadlock
B. there exist a safe sequence
C. all
D. none
Correct Answer: A. the system can allocate resources to each process in some order and still avoid a deadlock

Q74: Operating Systems (165)
The circular wait condition can be prevented by _____
A. defining a linear ordering of resource types
B. using thread
C. using pipes
D. all
Correct Answer: A. defining a linear ordering of resource types

Q75: Operating Systems (166)
A problem encountered in multitasking when a process is perpetually denied necessary resources is called _____
A. deadlock
B. starvation
C. inversion
D. aging
Correct Answer: B. starvation

Q76: Operating Systems (167)
To avoid deadlock _____
A. there must be a fixed number of resources to allocate

B. resource allocation must be done only once
C. all deadlocked processes must be aborted
D. inversion technique can be used
Correct Answer: A. there must be a fixed number of resources to allocate

Q77: Operating Systems (168)
The request and release of resources are _____
A. command line statements
B. interrupts
C. system calls
D. special programs
Correct Answer: C. system calls

Q78: Operating Systems (169)
What are Multithreaded programs?
A. lesser prone to deadlocks
B. more prone to deadlocks
C. not at all prone to deadlocks
D. none of the options
Correct Answer: B. more prone to deadlocks

Q79: Operating Systems (17)
The portion of the process scheduler in an operating system that dispatches processes is concerned with _____
A. assigning ready processes to waiting queue
B. assigning running processes to blocked queue
C. assigning ready processes to CPU
D. all
Correct Answer: C. assigning ready processes to CPU

Q80: Operating Systems (170)
For Mutual exclusion to prevail in the system _____
A. at least one resource must be held in a non sharable mode
B. the processor must be a uniprocessor rather than a multiprocessor
C. there must be at least one resource in a sharable mode
D. all
Correct Answer: A. at least one resource must be held in a non sharable mode

Q81: Operating Systems (171)
For a Hold and wait condition to prevail _____

A. A process must be not be holding a resource, but waiting for one to be freed, and then request to acquire it
B. A process must be holding at least one resource and waiting to acquire additional resources that are being held by other processes
C. A process must hold at least one resource and not be waiting to acquire additional resources
D. None
Correct Answer: B. A process must be holding at least one resource and waiting to acquire additional resources that are being held by other processes

Q82: Operating Systems (172)
Deadlock prevention is a set of methods _____
A. to ensure that at least one of the necessary conditions cannot hold
B. to ensure that all of the necessary conditions do not hold
C. to decide if the requested resources for a process have to be given or not
D. to recover from a deadlock
Correct Answer: A. to ensure that at least one of the necessary conditions cannot hold

Q83: Operating Systems (173)
For sharable resources, mutual exclusion _____
A. is required
B. is not required
C. may be or may not be required
D. none
Correct Answer: B. is not required

Q84: Operating Systems (174)
The disadvantage of a process being allocated all its resources before beginning its execution is _____
A. Low CPU utilization
B. Low resource utilization
C. Very high resource utilization
D. None
Correct Answer: B. Low resource utilization

Q85: Operating Systems (175)
To ensure no preemption, if a process is holding some resources and requests another resource that cannot be immediately allocated to it _____
A. then the process waits for the resources be allocated to it
B. the process keeps sending requests until the resource is allocated to it
C. the process resumes execution without the resource being allocated to it
D. then all resources currently being held are preempted

Correct Answer: D. then all resources currently being held are preempted

Q86: Operating Systems (176)
Each request requires that the system consider the _____ to decide whether the current request can be satisfied or must wait to avoid a future possible deadlock.
A. resources currently available
B. processes that have previously been in the system
C. resources currently allocated to each process
D. future requests and releases of each process
Correct Answer: A. resources currently available

Q87: Operating Systems (177)
Given a priori information about the _____ number of resources of each type that maybe requested for each process, it is possible to construct an algorithm that ensures that the system will never enter a deadlock state.
A. minimum
B. average
C. maximum
D. approximate
Correct Answer: C. maximum

Q88: Operating Systems (178)
A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.
A. resource allocation state
B. system storage state
C. operating system
D. resources
Correct Answer: A. resource allocation state

Q89: Operating Systems (179)
A system is in a safe state only if there exists a _____
A. safe allocation
B. safe resource
C. safe sequence
D. all
Correct Answer: C. safe sequence

Q90: Operating Systems (18)
The FCFS algorithm is particularly troublesome for _____
A. operating systems

B. multiprocessor systems
C. time sharing systems
D. multiprogramming systems
Correct Answer: D. multiprogramming systems

Q91: Operating Systems (180)
A system has 12 magnetic tape drives and 3 processes : P0, P1, and P2. Process P0 requires 10 tape drives, P1 requires 4 and P2 requires 9 tape drives.
Process
P0
P1
P2
Maximum needs (process-wise: P0 through P2 top to bottom)
10
4
9
Currently allocated (process-wise)
5
2
2
Which of the following sequence is a safe sequence?
A. P0, P1, P2
B. P1, P2, P0
C. P2, P0, P1
D. P1, P0, P2
Correct Answer: D. P1, P0, P2

Q92: Operating Systems (181)
The resource allocation graph is not applicable to a resource allocation system _____
A. with multiple instances of each resource type
B. with a single instance of each resource type
C. single & multiple instances of each resource type
D. none
Correct Answer: A. with multiple instances of each resource type

Q93: Operating Systems (182)
The Banker's algorithm is _____ than the resource allocation graph algorithm.
A. less efficient
B. more efficient
C. equal
D. none
Correct Answer: A. less efficient

Q94: Operating Systems (183)
A system with 5 processes P0 through P4 and three resource types A, B, C have A with 10 instances, B with 5 instances, and C with 7 instances. At time t0, the following snapshot has been taken:

Process
P0
P1
P2
P3
P4

Allocation (process-wise : P0 through P4 top TO bottom)

| A | B | C |
|---|---|---|
| 0 | 1 | 0 |
| 2 | 0 | 0 |
| 3 | 0 | 2 |
| 2 | 1 | 1 |
| 0 | 0 | 2 |

MAX (process-wise: P0 through P4 top TO bottom)

| A | B | C |
|---|---|---|
| 7 | 5 | 3 |
| 3 | 2 | 2 |
| 9 | 0 | 2 |
| 2 | 2 | 2 |
| 4 | 3 | 3 |

Available

| A | B | C |
|---|---|---|
| 3 | 3 | 2 |

The sequence  leads the system to _____

A. an unsafe state
B. a safe state
C. a protected state
D. a deadlock
Correct Answer: B. a safe state

Q95: Operating Systems (184)
The wait-for graph is a deadlock detection algorithm that is applicable when _____

A. all resources have a single instance
B. all resources have multiple instances
C. all resources have multiple instances
D. all
Correct Answer: A. all resources have a single instance

Q96: Operating Systems (185)
An edge from process Pi to Pj in a wait for graph indicates that _____
A. Pi is waiting for Pj to release a resource that Pi needs
B. Pi is waiting for Pj to release a resource that Pi needs
C. all resources have multiple instances
D. all
Correct Answer: A. Pi is waiting for Pj to release a resource that Pi needs

Q97: Operating Systems (186)
If the wait for graph contains a cycle _____
A. then a deadlock does not exist
B. then a deadlock exists
C. hen a deadlock exists
D. either deadlock exists or system is in a safe state
Correct Answer: B. then a deadlock exists

Q98: Operating Systems (187)
If deadlocks occur frequently, the detection algorithm must be invoked _____
A. rarely
B. frequently
C. rarely & frequently
D. none
Correct Answer: B. frequently

Q99: Operating Systems (19)
For an effective operating system, when to check for deadlock?
A. at fixed time intervals
B. every time a resource request is made
C. every time a resource request is made at fixed time intervals
D. none
Correct Answer: C. every time a resource request is made at fixed time intervals

Q100: Operating Systems (2)
What is the main function of the command interpreter?
A. to provide the interface between the API and application program
B. to handle the files in the operating system
C. to get and execute the next user-specified command
D. None
Correct Answer: C. to get and execute the next user-specified command

Q101: Operating Systems (20)
A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.
A. operating system
B. resources
C. system storage state
D. resource allocation state
Correct Answer: D. resource allocation state

Q102: Operating Systems (21)
Swapping _____ be done when a process has pending I/O, or has to execute I/O operations only into operating system buffers.
A. must never
B. maybe
C. can
D. must
Correct Answer: A. must never

Q103: Operating Systems (22)
The main memory accommodates _____
A. cpu
B. user processes
C. operating system
D. all
Correct Answer: C. operating system

Q104: Operating Systems (23)
The operating system is responsible for?
A. bad-block recovery
B. booting from disk
C. disk initialization
D. all
Correct Answer: D. all

Q105: Operating Systems (24)
The operating system and the other processes are protected from being modified by an already running process because _____
A. every address generated by the CPU is being checked against the relocation and limit registers
B. they have a protection algorithm

C. they are in different memory spaces
D. they are in different logical addresses
Correct Answer: A. every address generated by the CPU is being checked against the relocation and limit registers

Q106: Operating Systems (25)
Using transient code, _____ the size of the operating system during program execution.
A. maintains
B. changes
C. increases
D. decreases
Correct Answer: B. changes

Q107: Operating Systems (26)
The operating system maintains a _____ table that keeps track of how many frames have been allocated, how many are there, and how many are available.
A. memory
B. mapping
C. page
D. frame
Correct Answer: D. frame

Q108: Operating Systems (27)
To obtain better memory utilization, dynamic loading is used. With dynamic loading, a routine is not loaded until it is called. For implementing dynamic loading _____
A. special support from operating system is essential
B. special support from hardware is required
C. user programs can implement dynamic loading without any special support from hardware or operating system
D. special support from both hardware and operating system is essential
Correct Answer: C. user programs can implement dynamic loading without any special support from hardware or operating system

Q109: Operating Systems (28)
The _____ presents a uniform device-access interface to the I/O subsystem, much as system calls provide a standard interface between the application and the operating system.
A. Device drivers
B. I/O systems
C. Devices
D. Buses
Correct Answer: A. Device drivers

Q110: Operating Systems (29)
In real time operating system _____
A. process scheduling can be done only once
B. all processes have the same priority
C. kernel is not required
D. a task must be serviced by its deadline period
Correct Answer: D. a task must be serviced by its deadline period

Q111: Operating Systems (3)
In Operating Systems, which of the following is/are CPU scheduling algorithms?
A. Priority
B. Round Robin
C. Shortest Job First
D. All
Correct Answer: D. All

Q112: Operating Systems (30)
Hard real time operating system has _____ jitter than a soft real time operating system.
A. equal
B. more
C. less
D. none
Correct Answer: C. less

Q113: Operating Systems (31)
For real time operating systems, interrupt latency should be _____
A. zero
B. minimal
C. maximum
D. dependent on the scheduling
Correct Answer: B. minimal

Q114: Operating Systems (32)
Which one of the following is a real time operating system?
A. Windows CE
B. RTLinux
C. VxWorks
D. All
Correct Answer: D. All

Q115: Operating Systems (33)
The priority of a process will _____ if the scheduler assigns it a static priority.
A. depends on the operating system
B. change
C. remains unchanged
D. none
Correct Answer: C. remains unchanged

Q116: Operating Systems (34)
If the sum of the working set sizes increases, exceeding the total number of available frames _____
A. the operating system selects a process to suspend
B. the system crashes
C. then the process crashes
D. the memory overflows
Correct Answer: A. the operating system selects a process to suspend

Q117: Operating Systems (35)
The information about all files is kept in _____
A. operating system
B. separate directory structure
C. swap space
D. none
Correct Answer: B. separate directory structure

Q118: Operating Systems (36)
The operating system keeps a small table containing information about all open files called _____
A. file table
B. directory table
C. open-file table
D. system table
Correct Answer: C. open-file table

Q119: Operating Systems (37)
What will happen in the single level directory?
A. All files are contained in the same directory
B. All files are contained in different directories all at the same level
C. Depends on the operating system
D. None

Correct Answer: A. All files are contained in the same directory

Q120: Operating Systems (38)
The operating system _____ the links when traversing directory trees, to preserve the acyclic structure of the system.
A. deletes
B. considers
C. ignores
D. none
Correct Answer: C. ignores

Q121: Operating Systems (39)
On systems where there are multiple operating system, the decision to load a particular one is done by _____
A. process control block
B. file control block
C. boot loader
D. bootstrap
Correct Answer: C. boot loader

Q122: Operating Systems (4)
To access the services of the operating system, the interface is provided by the _____
A. Library
B. System Calls
C. Assembly instructions
D. API - Application Programming Interface
Correct Answer: B. System Calls

Q123: Operating Systems (40)
The _____ program initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system.
A. bootstrap
B. main
C. bootloader
D. rom
Correct Answer: A. bootstrap

Q124: Operating Systems (41)
Network operating system runs on _____
A. every system in the network

B. server
C. both server and every system in the network
D. none
Correct Answer: B. server

Q125: Operating Systems (42)
What are the types of distributed operating systems?
A. Zone based Operating system
B. Level based Operating system
C. Network Operating system
D. All
Correct Answer: C. Network Operating system

Q126: Operating Systems (43)
The systems which allow only one process execution at a time, are called _____
A. uniprogramming systems
B. uniprocessing systems
C. unitasking systems
D. none
Correct Answer: B. uniprocessing systems

Q127: Operating Systems (44)
In operating system, each process has its own _____
A. address space and global variables
B. open files
C. pending alarms, signals and signal handlers
D. all
Correct Answer: D. all

Q128: Operating Systems (45)
A process can be terminated due to _____
A. normal exit
B. fatal error
C. killed by another process
D. all
Correct Answer: D. all

Q129: Operating Systems (46)
What is the ready state of a process?
A. when process is scheduled to run after some execution
B. when process is unable to run until some task has been completed

C. when process is using the CPU
D. none
Correct Answer: A. when process is scheduled to run after some execution

Q130: Operating Systems (47)
What is interprocess communication?
A. communication within the process
B. communication between two process
C. communication between two threads of same process
D. none
Correct Answer: B. communication between two process

Q131: Operating Systems (48)
A set of processes is deadlock if _____
A. each process is blocked and will remain so forever
B. each process is terminated
C. all processes are trying to kill each other
D. None
Correct Answer: A. each process is blocked and will remain so forever

Q132: Operating Systems (49)
A process stack does not contain _____
A. Function parameters
B. Local variables
C. Return addresses
D. PID of child process
Correct Answer: D. PID of child process

Q133: Operating Systems (5)
CPU scheduling is the basis of _____
A. multiprogramming operating systems
B. larger memory sized systems
C. multiprocessor systems
D. none
Correct Answer: A. multiprogramming operating systems

Q134: Operating Systems (50)
Which system call can be used by a parent process to determine the termination of child process?
A. wait
B. exit

C. fork
D. get
Correct Answer: A. wait

Q135: Operating Systems (51)
The address of the next instruction to be executed by the current process is provided by the _____
A. CPU registers
B. Program counter
C. Process stack
D. Pipe
Correct Answer: B. Program counter

Q136: Operating Systems (52)
A Process Control Block(PCB) does not contain which of the following?
A. Code
B. Stack
C. Bootstrap Program
D. Data
Correct Answer: C. Bootstrap Program

Q137: Operating Systems (53)
The number of processes completed per unit time is known as _____
A. Output
B. Throughput
C. Efficiency
D. Capacity
Correct Answer: B. Throughput

Q138: Operating Systems (54)
The state of a process is defined by _____
A. the current activity of the process
B. the final activity of the process
C. the activity just executed by the process
D. the activity to next be executed by the process
Correct Answer: A. the current activity of the process

Q139: Operating Systems (55)
Which of the following is not the state of a process?
A. New
B. Old

C. Waiting
D. Running
Correct Answer: B. Old

Q140: Operating Systems (56)
What is a Process Control Block?
A. Process type variable
B. Data Structure
C. A secondary storage section
D. A Block in memory
Correct Answer: B. Data Structure

Q141: Operating Systems (57)
The entry of all the PCBs of the current processes is in _____
A. Process Register
B. Program Counter
C. Process Table
D. Process Unit
Correct Answer: C. Process Table

Q142: Operating Systems (58)
What is the degree of multiprogramming?
A. the number of processes executed per unit time
B. the number of processes in the ready queue
C. the number of processes in the I/O queue
D. the number of processes in memory
Correct Answer: D. the number of processes in memory

Q143: Operating Systems (59)
A single thread of control allows the process to perform _____
A. only one task at a time
B. multiple tasks at a time
C. only two tasks at a time
D. all
Correct Answer: A. only one task at a time

Q144: Operating Systems (6)
Which one of the following is not true?
A. kernel remains in the memory during the entire computer session
B. kernel is made of various modules which can not be loaded in running operating system

C. kernel is the first part of the operating system to load into memory during booting
D. kernel is the program that constitutes the central core of the operating system
Correct Answer: B. kernel is made of various modules which can not be loaded in running operating system

Q145: Operating Systems (60)
What is the objective of multiprogramming?
A. Have a process running at all time
B. Have multiple programs waiting in a queue ready to run
C. To increase CPU utilization
D. None
Correct Answer: C. To increase CPU utilization

Q146: Operating Systems (61)
Which of the following do not belong to queues for processes?
A. Job Queue
B. PCB Queue
C. Device Queue
D. Ready Queue
Correct Answer: B. PCB Queue

Q147: Operating Systems (62)
When the process issues an I/O request _____
A. It is placed in an I/O queue
B. It is placed in a waiting queue
C. It is placed in the ready queue
D. It is placed in the Job queue
Correct Answer: A. It is placed in an I/O queue

Q148: Operating Systems (63)
What will happen when a process terminates?
A. It is removed from all queues
B. It is removed from all, but the job queue
C. Its process control block is de-allocated
D. Its process control block is never de-allocated
Correct Answer: A. It is removed from all queues

Q149: Operating Systems (64)
What is a long-term scheduler?
A. It selects processes which have to be brought into the ready queue
B. It selects processes which have to be executed next and allocates CPU

C. It selects processes which heave to remove from memory by swapping
D. None
Correct Answer: A. It selects processes which have to be brought into the ready queue

Q150: Operating Systems (65)
If all processes I/O bound, the ready queue will almost always be _____ and the Short term Scheduler will have a _____ to do.
A. full, little
B. full, lot
C. empty, little
D. empty, lot
Correct Answer: C. empty, little

Q151: Operating Systems (66)
What is a medium-term scheduler?
A. It selects which process has to be brought into the ready queue
B. It selects which process has to be executed next and allocates CPU
C. It selects which process to remove from memory by swapping
D. None
Correct Answer: C. It selects which process to remove from memory by swapping

Q152: Operating Systems (67)
What is a short-term scheduler?
A. It selects which process has to be brought into the ready queue
B. It selects which process has to be executed next and allocates CPU
C. It selects which process to remove from memory by swapping
D. None
Correct Answer: B. It selects which process has to be executed next and allocates CPU

Q153: Operating Systems (68)
The primary distinction between the short term scheduler and the long term scheduler is _____
A. The length of their queues
B. The type of processes they schedule
C. The frequency of their execution
D. None
Correct Answer: C. The frequency of their execution

Q154: Operating Systems (69)
The only state transition that is initiated by the user process itself is _____
A. block

B. wakeup
C. dispatch
D. none
Correct Answer: A. block

Q155: Operating Systems (7)
Which one of the following errors will be handle by the operating system?
A. lack of paper in printer
B. connection failure in the network
C. power failure
D. all
Correct Answer: D. all

Q156: Operating Systems (70)
In a time-sharing operating system, when the time slot given to a process is completed, the process goes from the running state to the _____
A. Blocked state
B. Ready state
C. Suspended state
D. Suspended state
Correct Answer: B. Ready state

Q157: Operating Systems (71)
In a multiprogramming environment _____
A. the processor executes more than one process at a time
B. the programs are developed by more than one person
C. more than one process resides in the memory
D. a single user can execute many programs at the same time
Correct Answer: C. more than one process resides in the memory

Q158: Operating Systems (72)
Suppose that a process is in “Blocked” state waiting for some I/O service. When the service is completed, it goes to the _____
A. Running state
B. Ready state
C. Suspended state
D. Terminated state
Correct Answer: B. Ready state

Q159: Operating Systems (73)
The context of a process in the PCB of a process does not contain _____

A. the value of the CPU registers
B. the process state
C. memory-management information
D. context switch time
Correct Answer: D. context switch time

Q160: Operating Systems (74)
Which of the following need not necessarily be saved on a context switch between processes?
A. General purpose registers
B. Translation lookaside buffer
C. Program counter
D. All
Correct Answer: B. Translation lookaside buffer

Q161: Operating Systems (75)
Which of the following does not interrupt a running process?
A. A device
B. Timer
C. Schedular process
D. Power Failure
Correct Answer: C. Schedular process

Q162: Operating Systems (76)
Which process can be affected by other processes executing in the system?
A. cooperating process
B. child process
C. parent process
D. init process
Correct Answer: A. cooperating process

Q163: Operating Systems (77)
When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place is called _____
A. dynamic condition
B. race condition
C. essential condition
D. critical condition
Correct Answer: B. race condition

Q164: Operating Systems (78)
If a process is executing in its critical section, then no other processes can be executing in their critical section. What is this condition called?
A. mutual exclusion
B. critical exclusion
C. synchronous exclusion
D. asynchronous exclusion
Correct Answer: A. mutual exclusion

Q165: Operating Systems (79)
Which one of the following is a synchronization tool?
A. thread
B. pipe
C. semaphore
D. socket
Correct Answer: C. semaphore

Q166: Operating Systems (8)
Where is the operating system placed in the memory?
A. either low or high memory (depending on the location of interrupt vector)
B. in the low memory
C. in the high memory
D. none
Correct Answer: A. either low or high memory (depending on the location of interrupt vector)

Q167: Operating Systems (80)
A semaphore is a shared integer variable _____
A. that can not drop below zero
B. that can not be more than zero
C. that can not drop below one
D. that can not be more than one
Correct Answer: A. that can not drop below zero

Q168: Operating Systems (81)
Mutual exclusion can be provided by the _____
A. mutex locks
B. binary semaphores
C. both mutex locks and binary semaphores
D. none
Correct Answer: C. both mutex locks and binary semaphores

Q169: Operating Systems (82)
When high priority task is indirectly preempted by medium priority task effectively inverting the relative priority of the two tasks, the scenario is called _____
A. priority inversion
B. priority removal
C. priority exchange
D. priority modification
Correct Answer: A. priority inversion

Q170: Operating Systems (83)
Process synchronization can be done on _____
A. hardware level
B. software level
C. both hardware and software level
D. none
Correct Answer: C. both hardware and software level

Q171: Operating Systems (84)
A monitor is a module that encapsulates _____
A. shared data structures
B. procedures that operate on shared data structure
C. synchronization between concurrent procedure invocation
D. all
Correct Answer: D. all

Q172: Operating Systems (85)
To enable a process to wait within the monitor _____
A. a condition variable must be declared as condition
B. condition variables must be used as boolean objects
C. semaphore must be used
D. all
Correct Answer: A. a condition variable must be declared as condition

Q173: Operating Systems (86)
Restricting the child process to a subset of the parent's resources prevents any process from _____
A. overloading the system by using a lot of secondary storage
B. under-loading the system by very less CPU utilization
C. overloading the system by creating a lot of sub-processes
D. crashing the system by utilizing multiple resources

Correct Answer: C. overloading the system by creating a lot of sub-processes

Q174: Operating Systems (87)
A parent process calling _____ system call will be suspended until children processes terminate.
A. wait
B. fork
C. exit
D. exec
Correct Answer: A. wait

Q175: Operating Systems (88)
Cascading termination refers to termination of all child processes if the parent process terminates _____
A. Normally
B. Abnormally
C. Any one
D. None
Correct Answer: C. Any one

Q176: Operating Systems (89)
With _____ only one process can execute at a time; meanwhile all other process are waiting for the processor. With _____ more than one process can be running simultaneously each on a different processor.
A. Uniprogramming, Multiprogramming
B. Multiprocessing, Multiprogramming
C. Multiprogramming, Multiprocessing
D. Uniprogramming, Multiprocessing
Correct Answer: D. Uniprogramming, Multiprocessing

Q177: Operating Systems (9)
If a process fails, most operating system write the error information to a _____
A. new file
B. another running process
C. log file
D. none
Correct Answer: C. log file

Q178: Operating Systems (90)
In UNIX, the return value for the fork system call is _____ for the child process and _____ for the parent process.

A. A Negative integer, Zero
B. Zero, A Negative integer
C. Zero, A nonzero integer
D. A nonzero integer, Zero
Correct Answer: C. Zero, A nonzero integer

Q179: Operating Systems (91)
The child process can _____
A. be a duplicate of the parent process
B. never be a duplicate of the parent process
C. cannot have another program loaded into it
D. never have another program loaded into it
Correct Answer: A. be a duplicate of the parent process

Q180: Operating Systems (92)
The child process completes execution, but the parent keeps executing, then the child process is known as _____
A. Orphan
B. Zombie
C. Body
D. Dead
Correct Answer: B. Zombie

Q181: Operating Systems (93)
What is Interprocess communication?
A. allows processes to communicate and synchronize their actions when using the same address space
B. allows processes to communicate and synchronize their actions
C. allows the processes to only synchronize their actions without communication
D. none
Correct Answer: B. allows processes to communicate and synchronize their actions

Q182: Operating Systems (94)
Message passing system allows processes to _____
A. communicate with each other without sharing the same address space
B. communicate with one another by resorting to shared data
C. share data
D. name the recipient or sender of the message
Correct Answer: A. communicate with each other without sharing the same address space

Q183: Operating Systems (95)

Which of the following two operations are provided by the IPC facility?
A. write & delete message
B. delete & receive message
C. send & delete message
D. receive & send message
Correct Answer: D. receive & send message

Q184: Operating Systems (96)
Messages sent by a process _____
A. have to be of a fixed size
B. have to be a variable size
C. can be fixed or variable sized
D. none
Correct Answer: C. can be fixed or variable sized

Q185: Operating Systems (97)
The link between two processes P and Q to send and receive messages is called _____
A. communication link
B. message-passing link
C. synchronization link
D. all
Correct Answer: A. communication link

Q186: Operating Systems (98)
Which of the following are TRUE for direct communication?
A. A communication link can be associated with N number of process(N = max. number of processes supported by system)
B. A communication link is associated with exactly two processes
C. Exactly N/2 links exist between each pair of processes(N = max. number of processes supported by system)
D. Exactly two link exists between each pair of processes
Correct Answer: B. A communication link is associated with exactly two processes

Q187: Operating Systems (99)
In indirect communication between processes P and Q _____
A. there is another process R to handle and pass on the messages between P and Q
B. there is another machine between the two processes to help communication
C. there is a mailbox to help communication between P and Q
D. none
Correct Answer: C. there is a mailbox to help communication between P and Q